

Transformations and the Library of Relations (Part IV) Homework

The transformation properties can be used with each of the following pathways to graph the corresponding equations. But in most of these pathways, expressions must be converted into a different form in order to use the transformation properties in the way that they are written. Choose the pathways in which no modification of expressions is required. Then, use your chosen pathway to graph the corresponding equation.

① $-2y - 2 = |x + 3|$

Pathway #1

$$y = |x|$$

$$y - 2 = |x + 3|$$

$$2y - 2 = |x + 3|$$

$$2(-y) - 2 = |x + 3|$$

Pathway #2

$$y = |x|$$

$$y - 2 = |x + 3|$$

$$-y - 2 = |x + 3|$$

$$-2y - 2 = |x + 3|$$

Pathway #3

$$y = |x|$$

$$2y = |x|$$

$$2(-y) = |x|$$

$$-2y - 2 = |x + 3|$$

$$2(-y) = -2y$$

Pathway #4

$$y = |x|$$

$$2y = |x|$$

$$2y - 2 = |x + 3|$$

$$2(-y) - 2 = |x + 3|$$

Pathway #5

$$y = |x|$$

$$-y = |x|$$

$$-2y = |x|$$

$$-2y - 2 = |x + 3|$$

Pathway #6

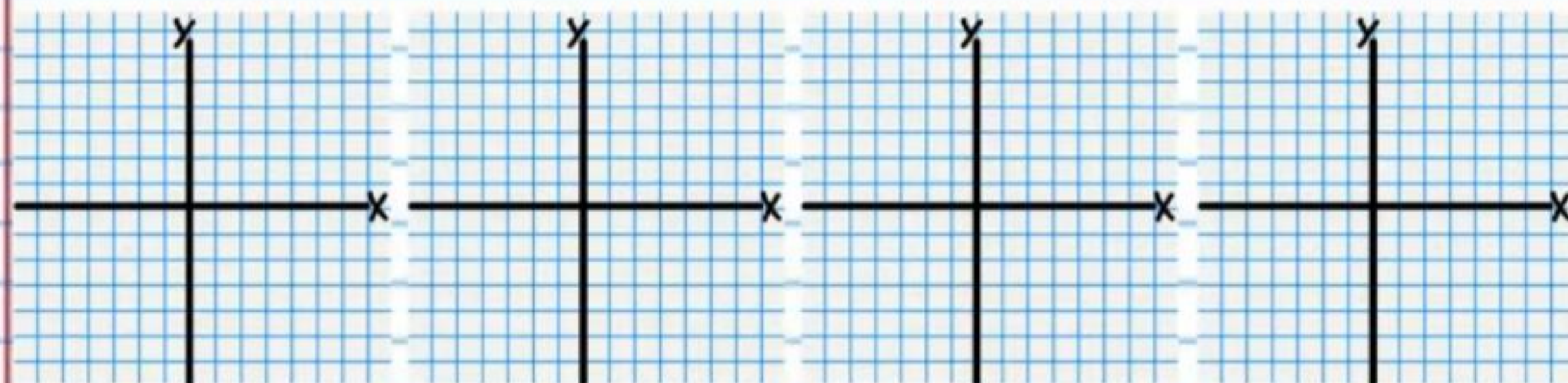
$$y = |x|$$

$$-y = |x|$$

$$-y - 2 = |x + 3|$$

$$-2y - 2 = |x + 3|$$

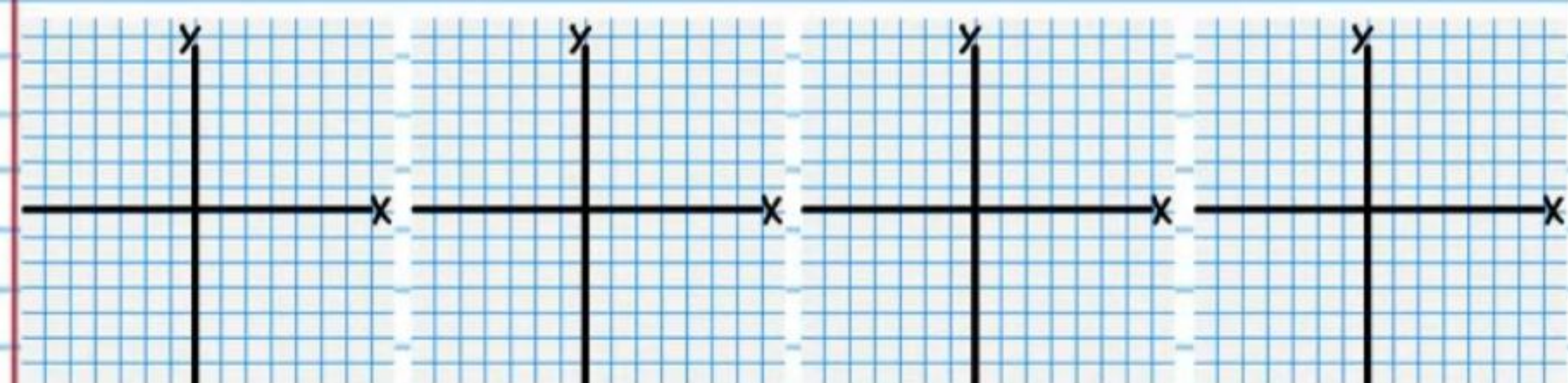
Parent Relation Transformation #1 Transformation #2 Final Transformation



$$\textcircled{2} -y-1 = \sqrt{3(x-2)}$$

Pathway #1	Pathway #2	Pathway #3
$y = \sqrt{x}$	$y = \sqrt{x}$	$y = \sqrt{x}$
$y-1 = \sqrt{x-2}$	$y-1 = \sqrt{x-2}$	$y = \sqrt{3x}$
$y-1 = \sqrt{3(x-2)}$	$-y-1 = \sqrt{x-2}$	$-y = \sqrt{3x}$
$-y-1 = \sqrt{3(x-2)}$	$-y-1 = \sqrt{3(x-2)}$	$-y-1 = \sqrt{3(x-2)}$
Pathway #4	Pathway #5	Pathway #6
$y = \sqrt{x}$	$y = \sqrt{x}$	$y = \sqrt{x}$
$y = \sqrt{3x}$	$-y = \sqrt{x}$	$-y = \sqrt{x}$
$y-1 = \sqrt{3(x-2)}$	$-y = \sqrt{3x}$	$-y-1 = \sqrt{x-2}$
$-y-1 = \sqrt{3(x-2)}$	$-y-1 = \sqrt{3(x-2)}$	$-y-1 = \sqrt{3(x-2)}$

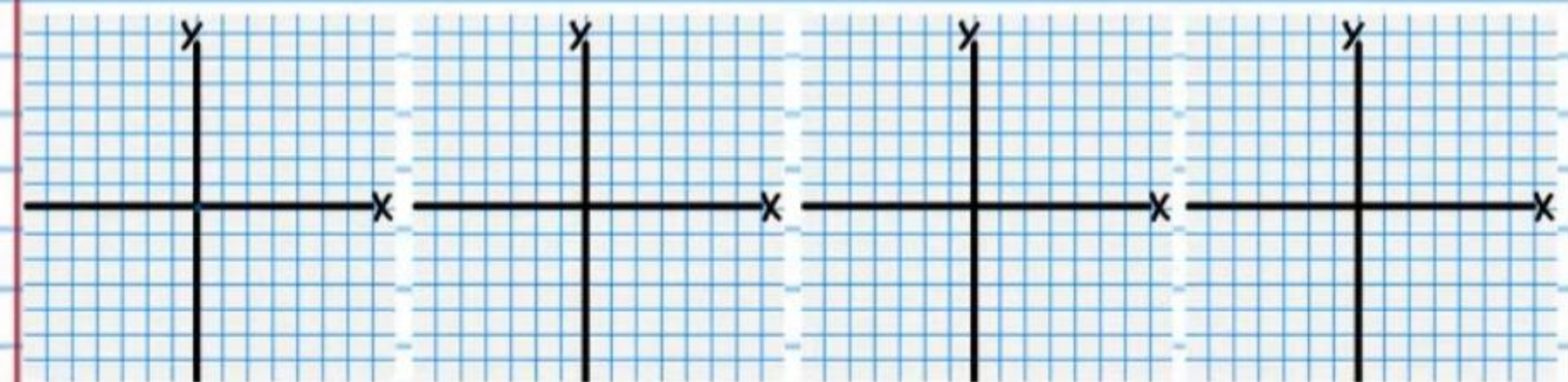
Parent Relation Transformation #1 Transformation #2 Final Transformation



$$\textcircled{3} y+3 = \left[\frac{-(x+1)}{2} \right]^2$$

Pathway #1	Pathway #2	Pathway #3
$y = x^2$	$y = x^2$	$y = x^2$
$y+3 = (x+1)^2$	$y+3 = (x+1)^2$	$y = \left(\frac{x}{2}\right)^2$
$y+3 = \left[\frac{x+1}{2}\right]^2$	$y+3 = [-(x+1)]^2$	$y = \left(\frac{-x}{2}\right)^2$
$y+3 = \left[\frac{-(x+1)}{2}\right]^2$	$y+3 = \left[\frac{-(x+1)}{2}\right]^2$	$y+3 = \left[\frac{-(x+1)}{2}\right]^2$
Pathway #4	Pathway #5	Pathway #6
$y = x^2$	$y = x^2$	$y = x^2$
$y = \left(\frac{x}{2}\right)^2$	$y = (-x)^2$	$y = (-x)^2$
$y+3 = \left(\frac{x+1}{2}\right)^2$	$y = \left(-\frac{x}{2}\right)^2$	$y+3 = [-(x+1)]^2$
$y+3 = \left[\frac{-(x+1)}{2}\right]^2$	$y+3 = \left[-\frac{x+1}{2}\right]^2$	$y+3 = \left[\frac{-(x+1)}{2}\right]^2$

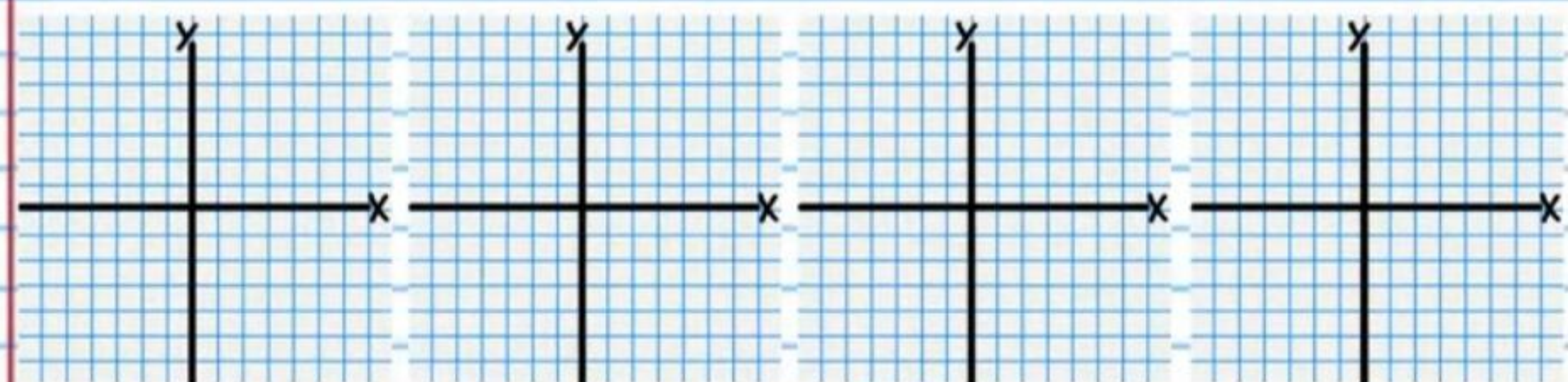
Parent Relation Transformation #1 Transformation #2 Final Transformation



$$\textcircled{4} -\frac{y}{3}-1=(x+1)^3$$

Pathway #1	Pathway #2	Pathway #3
$y=x^3$	$y=x^3$	$y=x^3$
$y-1=(x+1)^3$	$y-1=(x+1)^3$	$\frac{y}{3}=x^3$
$\frac{y}{3}-1=(x+1)^3$	$-y-1=(x+1)^3$	$-\frac{y}{3}=x^3$
$-\frac{y}{3}-1=(x+1)^3$	$-\frac{y}{3}-1=(x+1)^3$	$-\frac{y}{3}-1=(x+1)^3$
Pathway #4	Pathway #5	Pathway #6
$y=x^3$	$y=x^3$	$y=x^3$
$\frac{y}{3}=x^3$	$-y=x^3$	$-y=x^3$
$\frac{y}{3}-1=(x+1)^3$	$-\frac{y}{3}=x^3$	$-y-1=(x+1)^3$
$-\frac{y}{3}-1=(x+1)^3$	$-\frac{y}{3}-1=(x+1)^3$	$-\frac{y}{3}-1=(x+1)^3$

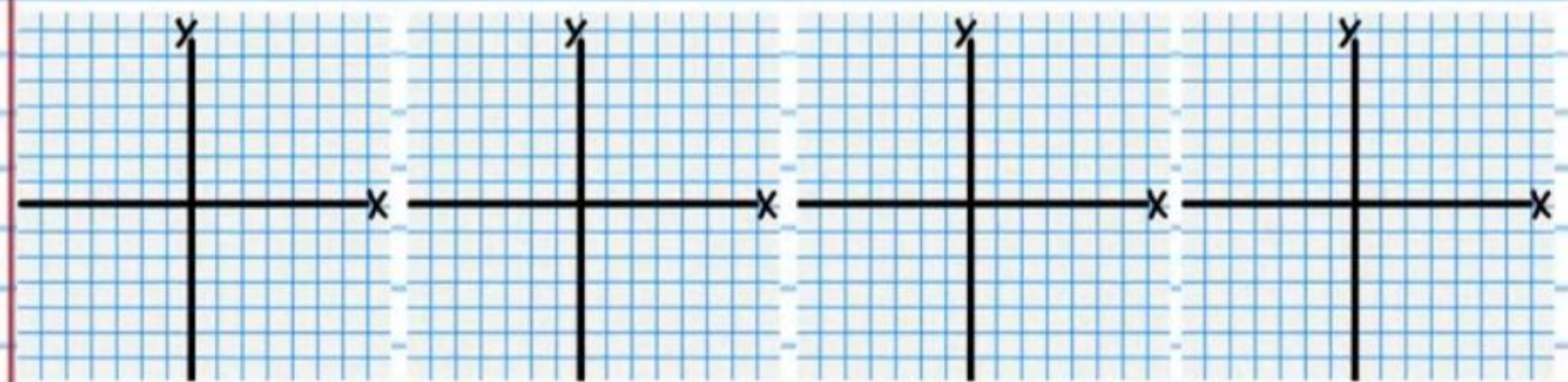
Parent Relation Transformation #1 Transformation #2 Final Transformation



$$\textcircled{5} \frac{y}{2}-2=3^{-(x-2)}$$

Pathway #1	Pathway #2	Pathway #3
$y=3^x$	$y=3^x$	$y=3^x$
$y-2=3^{x-2}$	$y-2=3^{x-2}$	$\frac{y}{2}=3^x$
$\frac{y}{2}-2=3^{x-2}$	$y-2=3^{-(x-2)}$	$\frac{y}{2}=3^{-x}$
$\frac{y}{2}-2=3^{-(x-2)}$	$\frac{y}{2}-2=3^{-(x-2)}$	$\frac{y}{2}-2=3^{-(x-2)}$
Pathway #4	Pathway #5	Pathway #6
$y=3^x$	$y=3^x$	$y=3^x$
$\frac{y}{2}=3^x$	$y=3^{-x}$	$y=3^{-x}$
$\frac{y}{2}-2=3^{x-2}$	$\frac{y}{2}=3^{-x}$	$y-2=3^{-(x-2)}$
$\frac{y}{2}-2=3^{-(x-2)}$	$\frac{y}{2}-2=3^{-(x-2)}$	$\frac{y}{2}-2=3^{-(x-2)}$

Parent Relation Transformation #1 Transformation #2 Final Transformation



Logarithm Property III

$$\log_b b = 1$$

Logarithm Property II

$$\log_b b^x = x$$

Logarithm Property IX

$$\log_b 1 = 0$$

$$\textcircled{6} -(y+2) = \log_3 \left(\frac{x}{2} + 1 \right)$$

Pathway #1

$$y = \log_3 x$$

$$y+2 = \log_3 (x+1)$$

$$y+2 = \log_3 \left(\frac{x}{2} + 1 \right)$$

$$-(y+2) = \log_3 \left(\frac{x}{2} + 1 \right)$$

Pathway #2

$$y = \log_3 x$$

$$y+2 = \log_3 (x+1)$$

$$-(y+2) = \log_3 (x+1)$$

$$-(y+2) = \log_3 \left(\frac{x}{2} + 1 \right)$$

Pathway #3

$$y = \log_3 x$$

$$y = \log_3 \left(\frac{x}{2} \right)$$

$$-y = \log_3 \left(\frac{x}{2} \right)$$

$$-(y+2) = \log_3 \left(\frac{x}{2} + 1 \right)$$

Pathway #4

$$y = \log_3 x$$

$$y = \log_3 \left(\frac{x}{2} \right)$$

$$y+2 = \log_3 \left(\frac{x}{2} + 1 \right)$$

$$-(y+2) = \log_3 \left(\frac{x}{2} + 1 \right)$$

Pathway #5

$$y = \log_3 x$$

$$-y = \log_3 x$$

$$-y = \log_3 \left(\frac{x}{2} \right)$$

$$-(y+2) = \log_3 \left(\frac{x}{2} + 1 \right)$$

Pathway #6

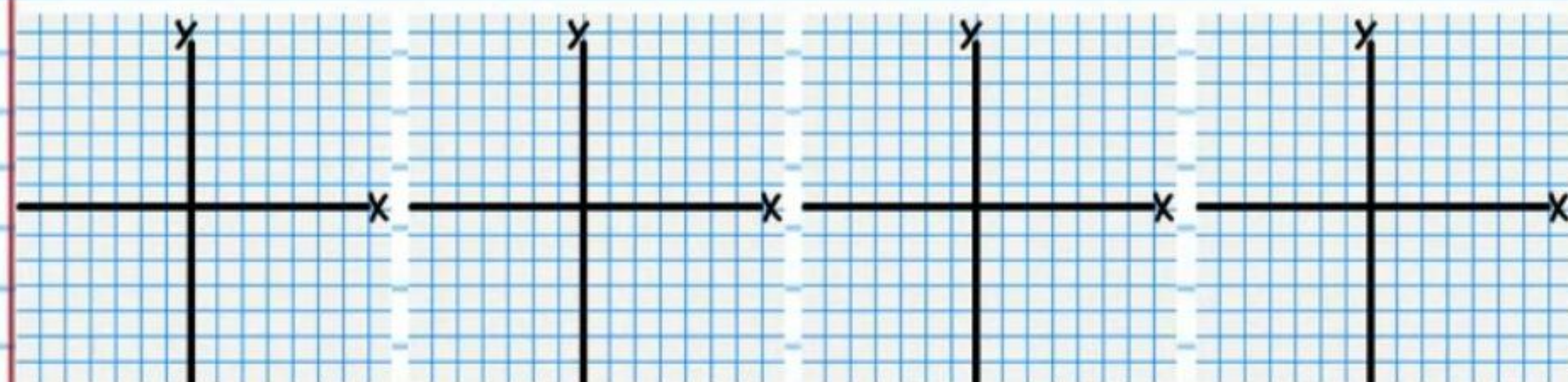
$$y = \log_3 x$$

$$-y = \log_3 x$$

$$-(y+2) = \log_3 (x+1)$$

$$-(y+2) = \log_3 \left(\frac{x}{2} + 1 \right)$$

Parent Relation Transformation #1 Transformation #2 Final Transformation



$$\textcircled{7} 2(y+3) = \frac{1}{-(x-1)}$$

Pathway #1

$$y = \frac{1}{x}$$

$$y+3 = \frac{1}{x-1}$$

$$2(y+3) = \frac{1}{x-1}$$

$$2(y+3) = \frac{1}{-(x-1)}$$

Pathway #2

$$y = \frac{1}{x}$$

$$y+3 = \frac{1}{x-1}$$

$$y+3 = \frac{1}{-(x-1)}$$

$$2(y+3) = \frac{1}{-(x-1)}$$

Pathway #3

$$y = \frac{1}{x}$$

$$2y = \frac{1}{x}$$

$$2y = \frac{1}{-x}$$

$$2(y+3) = \frac{1}{-(x-1)}$$

Pathway #4

$$y = \frac{1}{x}$$

$$2y = \frac{1}{x}$$

$$2(y+3) = \frac{1}{x-1}$$

$$2(y+3) = \frac{1}{-(x-1)}$$

Pathway #5

$$y = \frac{1}{x}$$

$$y = \frac{1}{-x}$$

$$2y = \frac{1}{-x}$$

$$2(y+3) = \frac{1}{-(x-1)}$$

Pathway #6

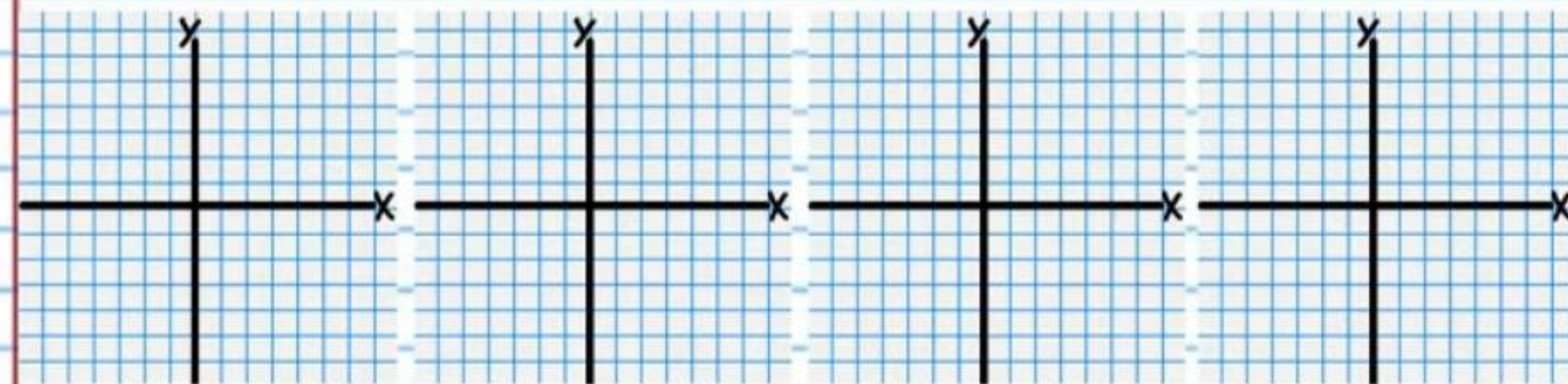
$$y = \frac{1}{x}$$

$$y = \frac{1}{-x}$$

$$y+3 = \frac{1}{-(x-1)}$$

$$2(y+3) = \frac{1}{-(x-1)}$$

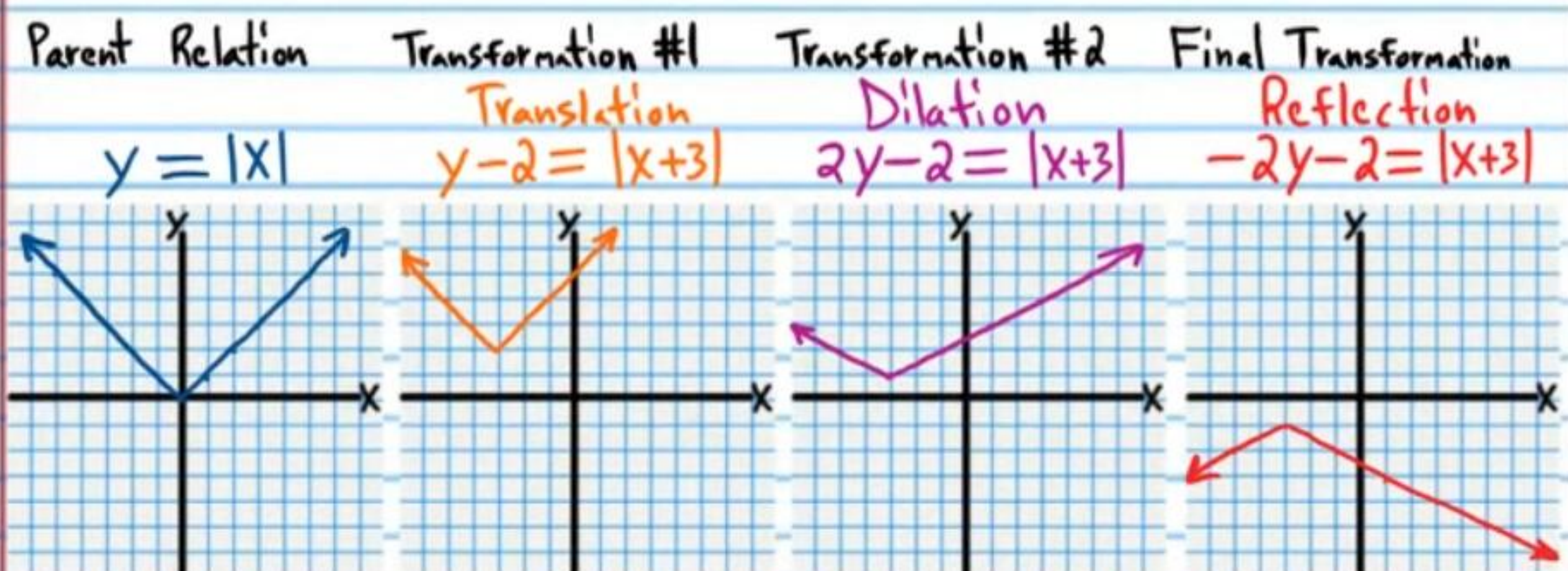
Parent Relation Transformation #1 Transformation #2 Final Transformation



Answers
① $-2y-2=|x+3|$

Pathway #1	Pathway #2	Pathway #3
$y= x $ ✓	$y= x $ ✓	$y= x $ ✓
$y-2= x+3 $ ✓	$y-2= x+3 $ ✓	$2y= x $ ✓
$2y-2= x+3 $ ✓	$-y-2= x+3 $ ✓	$2(-y)= x $ ✓
$2(-y)-2= x+3 $ ✓	$-2y-2= x+3 $ ✓	$-2y-2= x+3 $ ✗

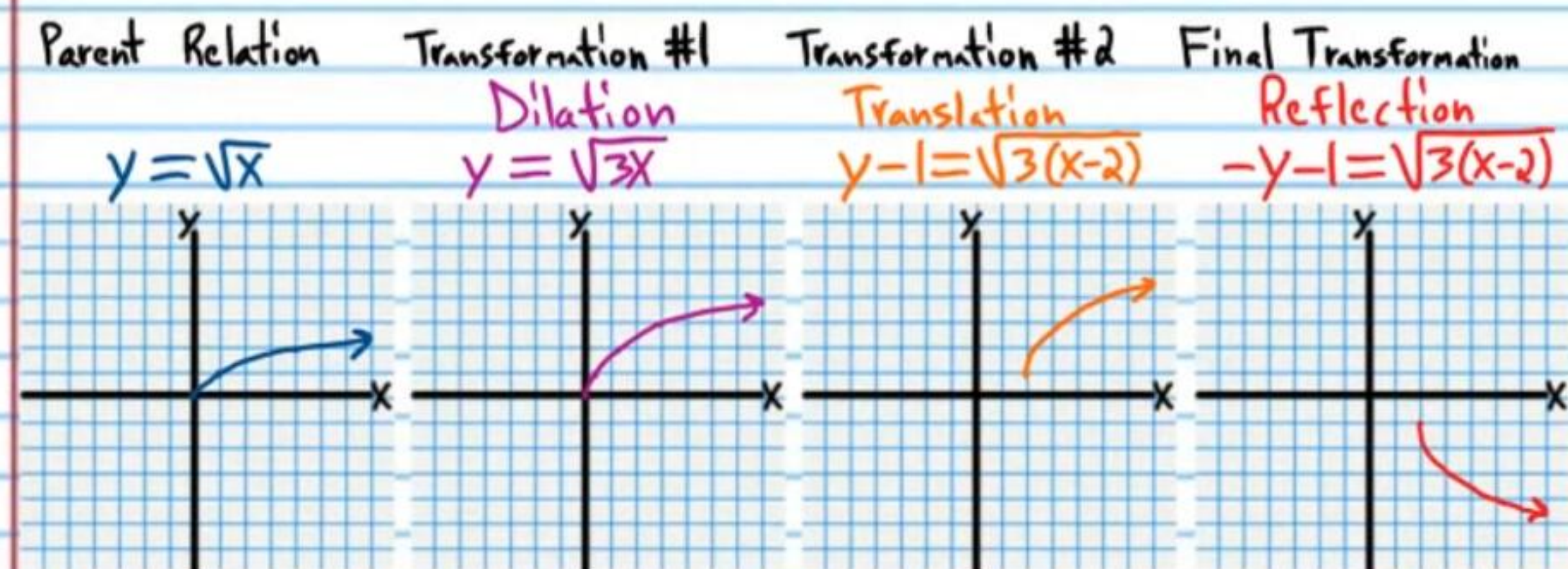
Pathway #4	Pathway #5	Pathway #6
$y= x $ ✓	$y= x $ ✓	$y= x $ ✓
$2y= x $ ✓	$-y= x $ ✓	$-y= x $ ✓
$2y-2= x+3 $ ✗	$-2y= x $ ✓	$-y-2= x+3 $ ✗
$2(-y)-2= x+3 $ ✓	$-2y-2= x+3 $ ✗	$-2y-2= x+3 $ ✓



② $-y-1=\sqrt{3(x-2)}$

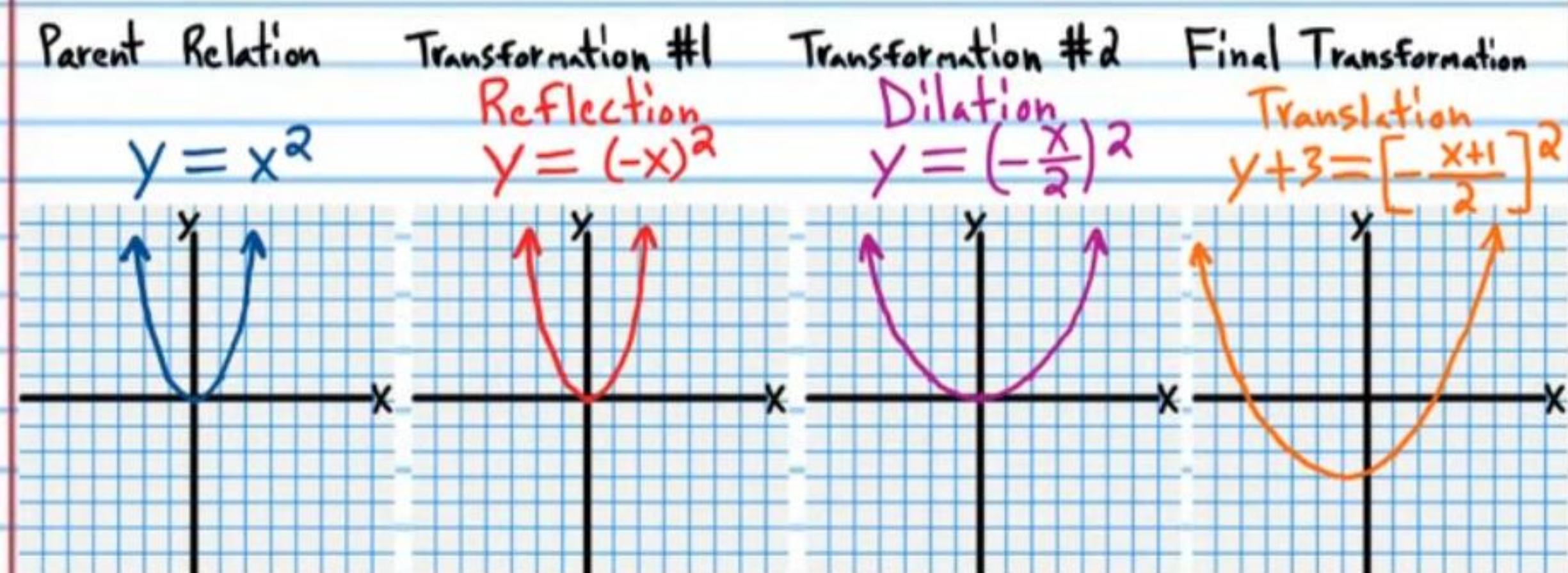
Pathway #1	Pathway #2	Pathway #3
$y=\sqrt{x}$ ✓	$y=\sqrt{x}$ ✓	$y=\sqrt{x}$ ✓
$y-1=\sqrt{x-2}$ ✓	$y-1=\sqrt{x-2}$ ✓	$y=\sqrt{3x}$ ✓
$y-1=\sqrt{3(x-2)}$ ✗	$-y-1=\sqrt{x-2}$ ✓	$-y=\sqrt{3x}$ ✓
$-y-1=\sqrt{3(x-2)}$ ✓	$-y-1=\sqrt{3(x-2)}$ ✗	$-y-1=\sqrt{3(x-2)}$ ✗

Pathway #4	Pathway #5	Pathway #6
$y=\sqrt{x}$ ✓	$y=\sqrt{x}$ ✓	$y=\sqrt{x}$ ✓
$y=\sqrt{3x}$ ✓	$-y=\sqrt{x}$ ✓	$-y=\sqrt{x}$ ✓
$y-1=\sqrt{3(x-2)}$ ✓	$-y=\sqrt{3x}$ ✓	$-y-1=\sqrt{x-2}$ ✗
$-y-1=\sqrt{3(x-2)}$ ✓	$-y-1=\sqrt{3(x-2)}$ ✗	$-y-1=\sqrt{3(x-2)}$ ✗



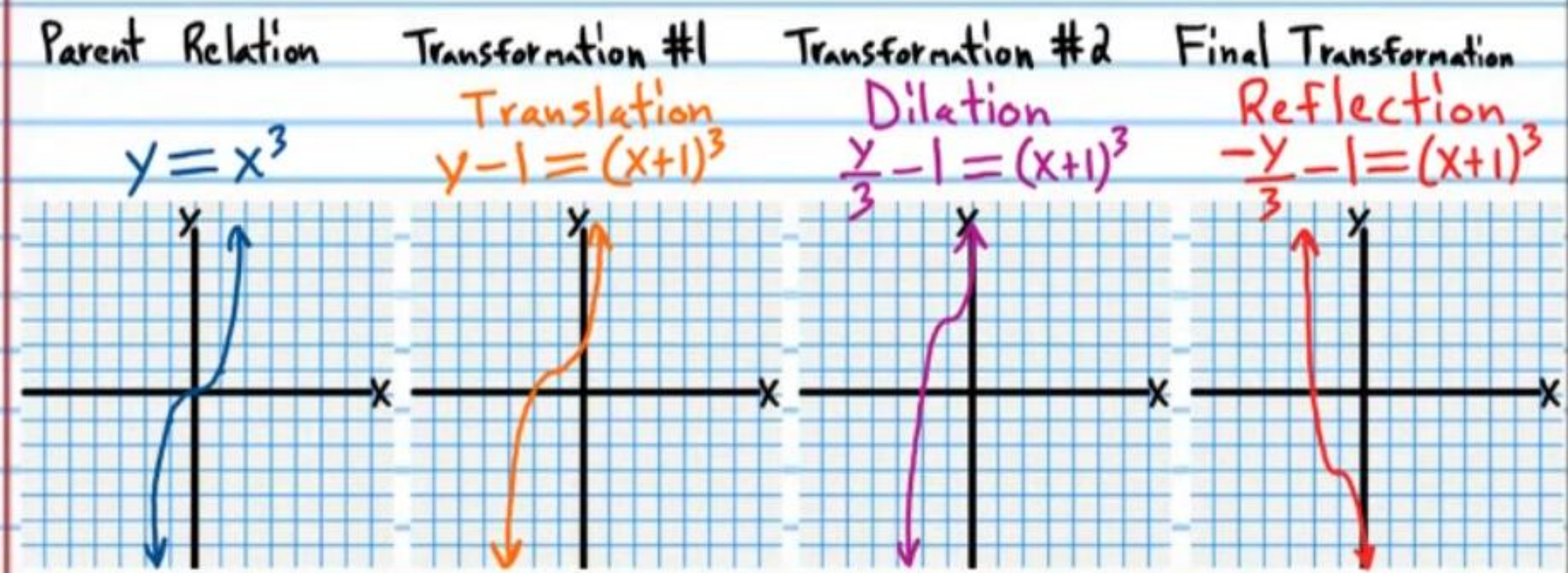
$$\textcircled{3} \ y+3 = \left[\frac{-(x+1)}{2} \right]^2$$

Pathway #1	Pathway #2	Pathway #3
$y = x^2$ ✓	$y = x^2$ ✓	$y = x^2$ ✓
$y+3 = (x+1)^2$ ✓	$y+3 = (x+1)^2$ ✓	$y = \left(\frac{x}{2}\right)^2$ ✓
$y+3 = \left[\frac{x+1}{2}\right]^2$ ✗	$y+3 = [-(x+1)]^2$ ✗	$y = \left(\frac{-x}{2}\right)^2$ ✓
$y+3 = \left[\frac{-(x+1)}{2}\right]^2$ ✗	$y+3 = \left[\frac{-(x+1)}{2}\right]^2$ ✗	$y+3 = \left[\frac{-(x+1)}{2}\right]^2$ ✓
Pathway #4	Pathway #5	Pathway #6
$y = x^2$ ✓	$y = x^2$ ✓	$y = x^2$ ✓
$y = \left(\frac{x}{2}\right)^2$ ✓	$y = (-x)^2$ ✓	$y = (-x)^2$ ✓
$y+3 = \left(\frac{x+1}{2}\right)^2$ ✓	$y = \left(-\frac{x}{2}\right)^2$ ✓	$y+3 = [-(x+1)]^2$ ✓
$y+3 = \left[\frac{-(x+1)}{2}\right]^2$ ✗	$y+3 = \left[-\frac{x+1}{2}\right]^2$ ✓	$y+3 = \left[\frac{-(x+1)}{2}\right]^2$ ✗



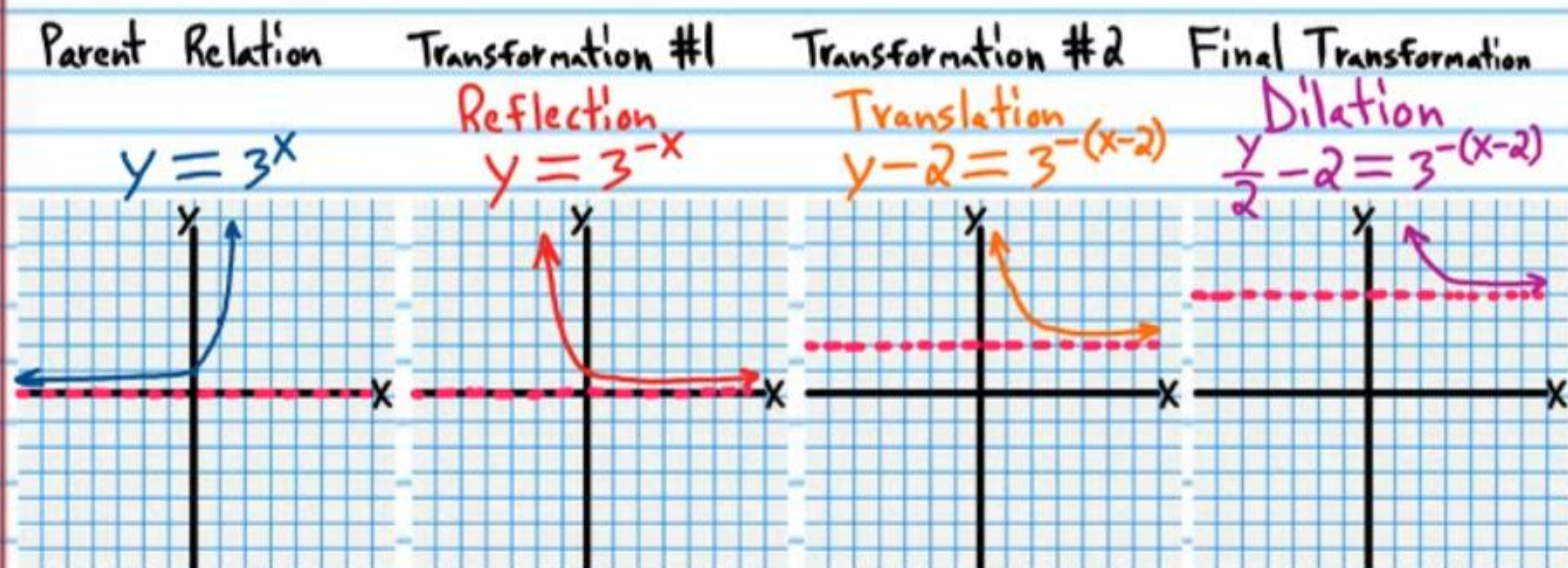
$$\textcircled{4} \ -\frac{y}{3} - 1 = (x+1)^3$$

Pathway #1	Pathway #2	Pathway #3
$y = x^3$ ✓	$y = x^3$ ✓	$y = x^3$ ✓
$y-1 = (x+1)^3$ ✓	$y-1 = (x+1)^3$ ✓	$\frac{y}{3} = x^3$ ✓
$\frac{y}{3} - 1 = (x+1)^3$ ✓	$-y-1 = (x+1)^3$ ✓	$-\frac{y}{3} = x^3$ ✓
$-\frac{y}{3} - 1 = (x+1)^3$ ✓	$-\frac{y}{3} - 1 = (x+1)^3$ ✓	$-\frac{y}{3} - 1 = (x+1)^3$ ✗
Pathway #4	Pathway #5	Pathway #6
$y = x^3$ ✓	$y = x^3$ ✓	$y = x^3$ ✓
$\frac{y}{3} = x^3$ ✓	$-y = x^3$ ✓	$-y = x^3$ ✓
$\frac{y}{3} - 1 = (x+1)^3$ ✗	$-\frac{y}{3} = x^3$ ✓	$-y-1 = (x+1)^3$ ✗
$-\frac{y}{3} - 1 = (x+1)^3$ ✓	$-\frac{y}{3} - 1 = (x+1)^3$ ✗	$-\frac{y}{3} - 1 = (x+1)^3$ ✓



$$\textcircled{5} \frac{y}{2} - 2 = 3^{-(x-2)}$$

Pathway #1	Pathway #2	Pathway #3
$y = 3^x$ ✓	$y = 3^x$ ✓	$y = 3^x$ ✓
$y - 2 = 3^{x-2}$ ✓	$y - 2 = 3^{x-2}$ ✓	$\frac{y}{2} = 3^x$ ✓
$\frac{y}{2} - 2 = 3^{x-2}$ ✓	$y - 2 = 3^{-(x-2)}$ ✗	$\frac{y}{2} = 3^{-x}$ ✓
$\frac{y}{2} - 2 = 3^{-(x-2)}$ ✗	$\frac{y}{2} - 2 = 3^{-(x-2)}$ ✓	$\frac{y}{2} - 2 = 3^{-(x-2)}$ ✗
Pathway #4	Pathway #5	Pathway #6
$y = 3^x$ ✓	$y = 3^x$ ✓	$y = 3^x$ ✓
$\frac{y}{2} = 3^x$ ✓	$y = 3^{-x}$ ✓	$y = 3^{-x}$ ✓
$\frac{y}{2} - 2 = 3^{x-2}$ ✗	$\frac{y}{2} = 3^{-x}$ ✓	$y - 2 = 3^{-(x-2)}$ ✓
$\frac{y}{2} - 2 = 3^{-(x-2)}$ ✗	$\frac{y}{2} - 2 = 3^{-(x-2)}$ ✗	$\frac{y}{2} - 2 = 3^{-(x-2)}$ ✓



Logarithm Property III

$$\log_b b = 1$$

Logarithm Property II

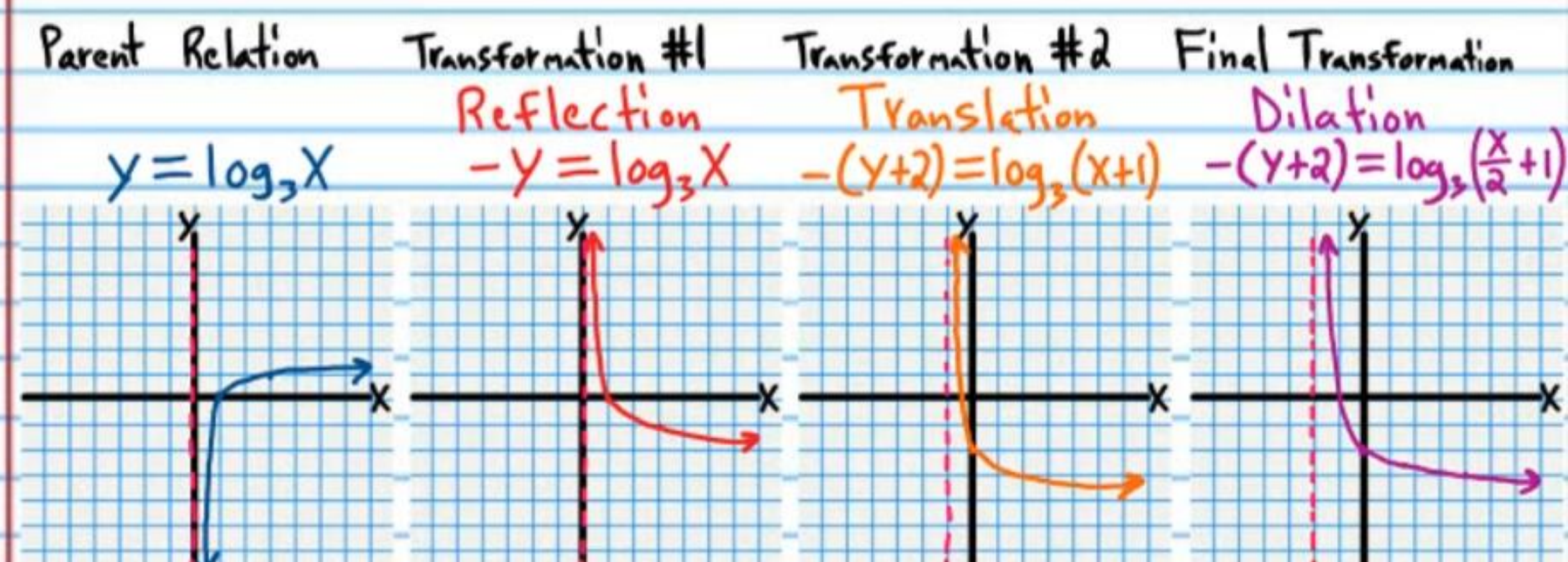
$$\log_b b^x = x$$

Logarithm Property IX

$$\log_b 1 = 0$$

$$\textcircled{6} -(y+2) = \log_3\left(\frac{x}{2} + 1\right)$$

Pathway #1	Pathway #2	Pathway #3
$y = \log_3 X$ ✓	$y = \log_3 X$ ✓	$y = \log_3 X$ ✓
$y + 2 = \log_3 (x + 1)$ ✓	$y + 2 = \log_3 (x + 1)$ ✓	$y = \log_3\left(\frac{x}{2}\right)$ ✓
$y + 2 = \log_3\left(\frac{x}{2} + 1\right)$ ✓	$-(y + 2) = \log_3 (x + 1)$ ✗	$-y = \log_3\left(\frac{x}{2}\right)$ ✓
$-(y + 2) = \log_3\left(\frac{x}{2} + 1\right)$ ✗	$-(y + 2) = \log_3\left(\frac{x}{2} + 1\right)$ ✓	$-(y + 2) = \log_3\left(\frac{x}{2} + 1\right)$ ✗
Pathway #4	Pathway #5	Pathway #6
$y = \log_3 X$ ✓	$y = \log_3 X$ ✓	$y = \log_3 X$ ✓
$y = \log_3\left(\frac{x}{2}\right)$ ✓	$-y = \log_3 X$ ✓	$-y = \log_3 X$ ✓
$y + 2 = \log_3\left(\frac{x}{2} + 1\right)$ ✗	$-y = \log_3\left(\frac{x}{2}\right)$ ✓	$-(y + 2) = \log_3 (x + 1)$ ✓
$-(y + 2) = \log_3\left(\frac{x}{2} + 1\right)$ ✗	$-(y + 2) = \log_3\left(\frac{x}{2} + 1\right)$ ✗	$-(y + 2) = \log_3\left(\frac{x}{2} + 1\right)$ ✓



$$\textcircled{7} 2(y+3) = \frac{1}{-(x-1)}$$

Pathway #1	Pathway #2	Pathway #3
$y = \frac{1}{x}$ ✓	$y = \frac{1}{x}$ ✓	$y = \frac{1}{x}$ ✓
$y+3 = \frac{1}{x-1}$ ✓	$y+3 = \frac{1}{x-1}$ ✓	$2y = \frac{1}{x}$ ✓
$2(y+3) = \frac{1}{x-1}$ ✗	$y+3 = \frac{1}{-(x-1)}$ ✗	$2y = \frac{1}{-x}$ ✓
$2(y+3) = \frac{1}{-(x-1)}$ ✗	$2(y+3) = \frac{1}{-(x-1)}$ ✗	$2(y+3) = \frac{1}{-(x-1)}$ ✓
Pathway #4	Pathway #5	Pathway #6
$y = \frac{1}{x}$ ✓	$y = \frac{1}{x}$ ✓	$y = \frac{1}{x}$ ✓
$2y = \frac{1}{x}$ ✓	$y = \frac{1}{-x}$ ✓	$y = \frac{1}{-x}$ ✓
$2(y+3) = \frac{1}{x-1}$ ✓	$2y = \frac{1}{-x}$ ✓	$y+3 = \frac{1}{-(x-1)}$ ✓
$2(y+3) = \frac{1}{-(x-1)}$ ✗	$2(y+3) = \frac{1}{-(x-1)}$ ✓	$2(y+3) = \frac{1}{-(x-1)}$ ✗

